

Scalability & Future Plan

Date	15 July 2026
Team ID	
Project Name	SkillEcho - Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High/Medium/Low)
1	Streamlit Cloud 1 GB RAM limit per instance	Restricts Whisper to the tiny (39M param) model; larger, more accurate models cannot be used	High
2	Single-threaded, synchronous execution (Python GIL)	Only one evaluation can be processed at a time per instance; concurrent submissions are serialised	High
3	No GPU access on Streamlit Community Cloud	All ML inference runs on CPU, adding ~15-25s latency per evaluation	Medium
4	Cold start latency on model load after deployment	First evaluation after restart takes 40-55s versus 15-25s once models are cached	Medium
5	Ephemeral disk storage for uploaded audio and PDFs	Uploaded audio files and generated PDFs are lost on instance reboot	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	1 concurrent evaluation (GIL-serialised, single Streamlit instance)	Introduce a Celery + Redis async task queue so 10-50+ evaluations can be processed concurrently by scaled workers
2	Data Storage	Ephemeral local disk for uploaded audio and generated PDFs	Migrate to AWS S3 / Google Cloud Storage for persistent, durable object storage
3	Performance	15-25s warm-cache / 40-55s cold-start response time on CPU-only instances	Move AI inference to GPU-backed Hugging Face Spaces (T4/A10G) to cut response time to 5-10s
4	Security	SHA-256 password hashing with session-based role checks	Adopt JWT token-based authentication with OAuth2 SSO (Google/GitHub) for production-grade security

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Celery + Redis async task queue for concurrent evaluations	2 weeks	Critical - unlocks concurrency beyond 1 user
Phase 2	JWT token-based authentication	1 week	Critical - production-grade security
Phase 3	Hugging Face Spaces GPU deployment (larger Whisper models)	1 week	High - faster response, better accuracy
Phase 3	S3 / GCS persistent file storage for audio and PDFs	1 week	High - durable storage across reboots
Phase 3	Multilingual Whisper support	3 days	High - expands to non-English institutions
Phase 4	LLM-powered qualitative feedback, React/Next.js frontend, comparative analytics dashboard, real-time streaming transcription, OAuth2 SSO	2-4 weeks each	Medium to Low - pedagogical value, mobile support, UX and convenience improvements